Examiner only

12. *(a)* Some students were given a mixture made up of about 50 % each of sodium chloride and sodium iodide and asked to design tests to show the presence and concentration of iodide ions in this mixture.

(i) One student decided to try electrolysis. He dissolved some of the mixture in distilled water to give a concentrated solution. He then passed electricity through the solution, using inert electrodes. The negative ions moved to the anode where they lost electrons.

State what was seen at the anode to confirm the presence of iodide ions.
Explain your answer, including a half-equation. [3]

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(ii) Another student decided to use a chemical test. She dissolved some of the mixture in distilled water and then added a little aqueous silver nitrate. After noting what was seen, she added some aqueous ammonia to the mixture and shook it.

State and explain what occurred during this test. [3]

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(iii) In a further test the concentration of the iodide ions present was found by a titration with potassium iodate(V) dissolved in a strong acid. 11.24 g of the mixture of sodium chloride and sodium iodide were dissolved in distilled water and made up to 250 cm^3. The iodide ions present in 25.0 cm^3 of this solution required 18.00 cm^3 of the potassium iodate(V) solution of concentration 0.100 mol dm^{-3}, to completely react.

Calculate the number of moles of iodide ions (present as sodium iodide, M_r 150) in 25.0 cm^3 of the solution and hence the exact percentage of sodium iodide in the mixture. [3]

(1 mol of potassium iodate(V) reacts with 2 mol of sodium iodide)

Percentage of sodium iodide = ... %

(b) Hydrogen chloride is a colourless gas that absorbs at 278 nm in the ultraviolet region of the electromagnetic spectrum.

Use this information and the data sheet to show that the energy of the H–Cl bond is 431 kJ mol^{-1}. [2]

Examiner only

(c) The table shows bond energy values and absorption maxima for the hydrogen halides.

H–X	Bond energy / kJ mol^{-1}	Absorption maximum / nm
H–F	562	212
H–Cl	431	278
H–Br	366	326
H–I	299	400
H–At		

The accepted range for wavelengths for the visible region of the electromagnetic spectrum is 400-700 nm.

Suggest and explain why it is probable that H–At would be a coloured gas. [2]

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13

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL – NEW

2410U10-1

S17-2410U10-1

CHEMISTRY – AS unit 1
The Language of Chemistry, Structure of Matter and Simple Reactions

FRIDAY, 26 MAY 2017 – MORNING

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 5.**	**10**	
Section B	**6.**	**20**	
	7.	**15**	
	8.	**17**	
	9.	**18**	
	Total	**80**	

2410U101 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.9**(*a*).

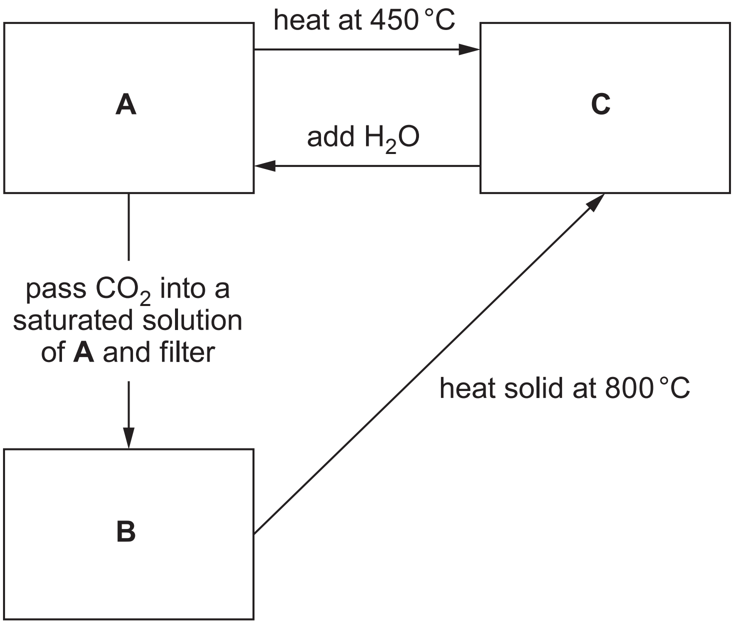
If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

MAY172410U10101

Examiner only

12. *(a)* A series of experiments is performed on three white solids, **A**, **B** and **C**. All contain the same Group 2 metal ion. Compound **A** is the metal hydroxide.

The experiment is summarised below.



- The compounds give a definite colour in a flame test
- Compound **A** is only slightly soluble in water
- Compound **B** is insoluble in water

(i) A student correctly identified the metal ion as calcium. Give **two** reasons why she came to that conclusion. [2]

....

....

....

....

(ii) Name compound **B**. [1]

....

Examiner only

(iii) 0.110 g of compound **C** was completely neutralised by 26.10 cm^3 of 0.150 mol dm^{-3} hydrochloric acid.

Given that 1 mol of compound **C** reacts with 2 mol of acid, calculate the relative formula mass of compound **C** and hence confirm that the metal ion is calcium.

You **must** show your working. [2]

(b) Group 2 metals are not the only elements that form 2+ ions.

A sample of an element has two isotopes, one with 70 neutrons and the other 72 neutrons. This element forms a 2+ ion containing 48 electrons. The relative abundances of the isotopes are 57.9 % and 42.1 % respectively.

Calculate the relative atomic mass of the element. Give your answer to **four** significant figures. [3]

A_r = ...

Examiner only

(c) Describe, giving **brief** experimental details, how you would obtain the maximum amount of magnesium carbonate from a known mass of magnesium, in a **two-stage** process. Include appropriate chemical / ionic equations in your answer. [6 QER]

14

Examiner only

12. A student was asked to find the identity of a Group 1 metal carbonate by titration.

He was told to use the following method.

- Weigh a sample of the carbonate in a weighing bottle.
- Transfer the carbonate into a beaker and weigh the bottle afterwards.
- Add water to the beaker to dissolve the carbonate.
- Transfer the solution to a volumetric flask.
- Add more water to make the final volume 250.0 cm^3 of solution.
- Accurately transfer 25.0 cm^3 of this solution into a conical flask.
- Add 2–3 drops of a suitable indicator to this solution.
- Fill a burette with 0.100 $mol\,dm^{-3}$ hydrochloric acid solution.
- Carry out a rough titration of the carbonate solution with the hydrochloric acid.
- Accurately repeat the titration until you get concordant titres and calculate a mean titre.

(a) Another student said that there were two errors in making the 250.0 cm^3 carbonate solution.

Error 1: A small amount of solid remained in the weighing bottle.

Error 2: A small amount of solution remained in the beaker.

Comment on the suggested errors.
If the student is correct suggest how the method could be improved.
If the student is incorrect, explain why. [2]

Error 1 ....

....

....

Error 2 ....

....

....

(b) State why he adds an indicator to this solution. [1]

....

....

(c) Suggest why he was told to carry out a rough titration first. [1]

....

....

Examiner only

(d) State what you understand by the term 'concordant titres'. [1]

....

....

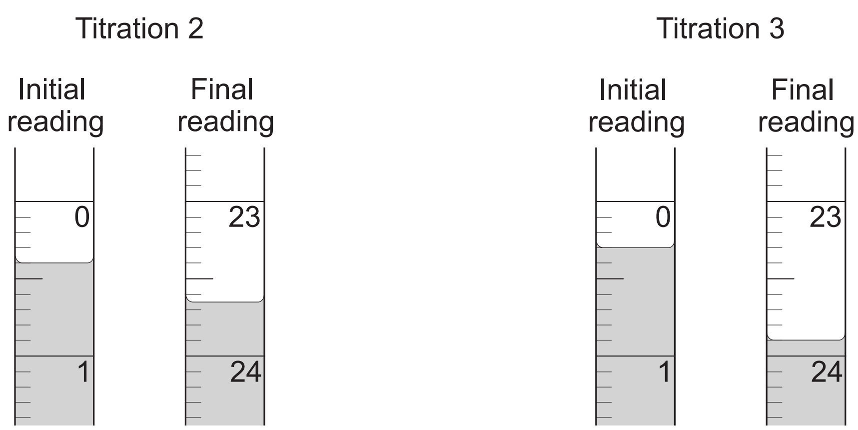
(e) Some of the student's results are shown below:

Mass of weighing bottle + carbonate / g	13.73
Mass of weighing bottle / g	12.48

Titration	Rough	1	2	3
Final reading / cm^3	24.20	23.70	....	....
Initial reading / cm^3	0.00	0.10	....	....
Titre / cm^3	....	....	....	....

concentration of hydrochloric acid = 0.100 mol dm^{-3}

The diagrams below show the initial burette reading and the final burette reading for the second and third titrations.



Complete the student's table and calculate the mean titre. [3]

mean titre = ... cm^3

Examiner only

(f) The equation for the reaction between the metal carbonate and hydrochloric acid is given below. **M** represents the symbol of the Group 1 metal.

$$\mathbf{M}_2CO_3 \quad + \quad 2HCl \longrightarrow 2\mathbf{M}Cl \quad + \quad H_2O \quad + \quad CO_2$$

(i) Calculate the number of moles of M_2CO_3 in $25.0\,cm^3$ of the solution. [2]

number of moles = ...

(ii) Calculate the relative formula mass of the carbonate and hence deduce the Group 1 metal in the carbonate.

You **must** show your working. [4]

group 1 metal = ...

14

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

S23-2410U10-1

2410U10-1

TUESDAY, 16 MAY 2023 – MORNING

CHEMISTRY – AS unit 1

The Language of Chemistry, Structure of Matter and Simple Reactions

1 hour 30 minutes

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1. to 6.	10	
Section B	7.	10	
	8.	12	
	9.	12	
	10.	15	
	11.	21	
	Total	80	

2410U101
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ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.7**(a).

BE*(X21-2410U10-1)